



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: Final Year B. Tech.	Subject: Cryptography and Network Security	Subject Code: PCC-IT401
Date: 19.08.2025	Time: 11:00 to 11:45 am	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Use of non-programmable calculator is permitted.

Q.N.		Marks	CO	BT
Q.1	a) Encrypt and Decrypt the plaintext "ATTACK IS TONIGHT" with the help of Hill cipher using the key $K = \begin{pmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{pmatrix}$	10 M	CO2	BT5
Q.2	a) Answer the following questions using a Playfair cipher is given by its key K= LARKSPUR. i) Encrypt the plaintext "SUNNY MOUNTAIN MEADOW" ii) Decrypt the cryptotext "ROCKY MOUNTAIN MEADOW"	05 M	CO1	BT3
	b) Explain Active attack and Passive attack with suitable example?	05 M	CO1	BT2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO1	Understand the concepts of classical encryption techniques and concepts of finite fields and number theory.
CO2	Explore working principles and utilities of various cryptographic algorithms including secret key cryptography, hashes and message digests, and public key algorithms.
CO3	Explore design issues and working principles of various authentication protocols, PKI standards.
CO4	Explore various secure communication standards including Kerberos, IPsec, and SSL/TLS and email.
CO5	Use existing cryptographic utilities to build programs for secure communication.